

# An Analysis Comparing Traditional and Digital Marketing (Advertising) using Chi Square test and Linear Regression model

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**Abstract** - In a market, two parties can come together to exchange products and services. The firms have adopted various techniques to push their product in the market; one of the recent techniques to sell their product by way of digital marketing. The advancement of technology has pushed individuals and communities to embrace digital technology and importance of online involvement has been elevated alongside that of conventional offline engagement, resulting in a marketing 4.0 combination of the two utilise it to keep track of and simplify their daily lives. This shift has paved the way for the incorporation of technological effects into company marketing strategies. The. Today's firms have recognised the importance of the digital environment in marketing to clients. This study's primary goal is to compare traditional marketing and digital marketing. It is compared using Statistical tools (i.e) Chi-square test and regression coefficients. This research focuses on how digital marketing is succeeding over traditional marketing. In this research we used both primary and secondary data.

**Keywords** - Marketing, Traditional Marketing, Digital Marketing, Techniques, Chi-square test, regression coefficients.

## I. INTRODUCTION

Marketing is any activity a business engages in to draw customers to its high-caliber messaging that promoting its goods and services. Marketing's goal is to use content to offer distinctive value to potential customers and customers in order to promote brand loyalty, show the worth of the product, and ultimately boost sales.

The practise of connecting consumers and offering products and services through digital platforms is referred to as "digital marketing." Any marketing activity that isn't carried out online is included. It is a conventional kind of marketing that connects with a semi-targeted audience by utilising a number of offline advertising and promotional strategies. Traditional marketing has evolved over time, but its fundamental elements have not. The providing of values

that people can regularly rely on was emphasised in traditional marketing in order to win the public's trust.



Support for project coordination was studied in publications of Arunprakash.N[1], Nazimsha.S[2], Pankaj Kumar Bharti[3], Raj Sinha[4], Sanjay Bhayani[5] and Sudha Lawrence[6]. The fields of traditional marketing and digital marketing have received extensive study. They emphasised the elements that have an impact on consumers' purchasing behaviours when it comes to traditional and digital marketing and claimed that the customer response to digital marketing is outstanding. To determine consumer preferences for traditional and digital marketing, more information is becoming available for consumers to utilise while searching for products and making purchases. Additionally, to determine whether technology, from print to digital media, revolutionises all sectors, including marketing. The widespread use of social media and the digital marketing has exploded in popularity, and the internet is directly responsible. This kind of marketing makes use of websites, mobile devices, social media, search engines, and other channels that serve a similar purpose. Digital marketing is frequently regarded as a cutting-edge tactic for companies to communicate with clients and understand client behaviour. It shares many concepts with conventional marketing. One kind of promotion is conventional marketing. In addition to examining the effectiveness and efficiency of digital marketing, this study aims to ascertain consumer preferences for traditional and digital marketing. In addition, the importance of both

traditional and digital marketing will be investigated and also determines how social media platforms are used in digital marketing.

#### *A. Purpose of the Study*

##### *a) Main goals:*

- To compare traditional and digital marketing.

##### *b) Secondary goals:*

- To study the customer preference towards traditional marketing and digital marketing
- To find on which platforms customers prefer to see advertisements.
- To determine the efficiency and effectiveness of traditional and digital marketing.

#### *B. Problem statement*

- Many people are unaware of digital marketing.
- You may be targeting the wrong audience by spending most of your digital marketing on underperforming channels.
- In contrast to traditional marketing, which is currently losing ground in the wake of the pandemic, digital marketing is expected to take over in the next years.
- Traditional marketing can be more costly than online marketing.

## II. RESEARCH METHODOLOGY

#### *A. Research*

Research is the methodical, scientific search for knowledge on a given subject or field. Research Techniques Survey research has been utilized to confirm the general public's attitude in light of the need, nature, and goals of this research project. The present study on "An analysis Comparing Traditional Marketing and Digital Marketing (Advertising) using chi square test and linear regression model" is based on a descriptive study since it depicts the state of the situation at the time, the attitudes, and the development.

#### *B. Research Design*

This study is based on descriptive research. The descriptive research that underpins this study. This kind of research is done to describe the features of the population. gathers information that is used to respond to a variety of "what, when, and how" queries about a specific community or group.

#### *C. Data Collection*

Data is defined as the information about something that is transformed in a form to efficiently move it from one place to another. Types of data are collected in this research are: **a) Primary Data** **b) Secondary Data**.

*a) Primary Data:* Information or data that is acquired directly from the research's focus target is referred to as "primary data." We directly gathered data by dispersing organized questionnaires.

*b) Secondary Data:* Information obtained from a variety of sources, including books, articles, and journals, is referred to as secondary data.

#### *D. Research Instruments*

The questionnaire was framed on the basis of objectives and distributed to people and the responses were collected. The questionnaires contained structured are closed questions which required the respondents to choose. Questions include Multi-choice questions and 5 point likert scale. Multi-choice and 5 point likert scale questions contains a list of number of answers and permit the respondentsto select the answer that seems approximates to their own choice.

#### *E. Sampling Techniques*

Sampling is a technique for selecting particular people or groups from a society in order to calculate demographic characteristics and derive statistical conclusions. Probability sampling and non-probability sampling are the two different categories of sampling procedures. We employed convenience non-probability sampling in this investigation.

*Sample Size:* There were 110 respondents in this study,

#### *F. Analysis Tool*

SPSS - is the programme used to compute or analyse the data. Statistical Package for the Social Sciences is known as SPSS. SPSS is a programme for data editing and analysis. All commonly used file formats for structured data, including Excel and plain text files, can be opened by SPSS. In this study we used Chi – Square & Linear regression for analyzing.

### G. Limitations of the Study

- As the research topic itself states the research work is mainly concerned on traditional and digital marketing.
- This study's scope has been restricted to a survey of Chennai District residents due to time and resource restrictions.
- The survey sample is limited only to - respondents Research work carried over is only in the urban areas.
- Almost all respondents are literates and no illiterate people were taken as samples.
- The respondents view and opinion may hold good for time being and may vary in future.

## III. ANALYSIS OF THE STUDY

### A. Statistical Analysis:

a) *Chi – Square*: Nonparametric chi-square tests are used for two distinct purposes. Test the idea that no association exists between two or more groups, populations, or criteria. (b) Evaluate how closely the observed and expected data distributions match. It is applied to the study of categorical data. It is not meant to analyse continuous or parametric data.

### b) Assumptions Underlying the Chi-Square Test

- Data is randomly sampled from the population
- A random sample of the population's data is taken.
- A cell value is considered fair if the expected count is greater than 5 and there are no cells with a count of zero.
- Sufficiently many samples. Chi-square testing on small samples could include Type II errors (that is, accepting a null hypothesis that is actually false). No projected maximum sample size exists. However, the basic minimum sample size is between 20 and 50.
- The variables taken into account must be mutually exclusive. This indicates that a variable cannot appear in more than one category and can only be tallied once in that category. Thus, nothing is counted more than once.

The creation of a 2 X 2 contingency table is the fundamental step in computing the chi-square test. [(corresponding row sum \* corresponding column sum) / total count] is a general formula for determining the expected count given the observed count. Degrees of freedom (df) are equal to (number of columns divided by 1) and (number of rows divided by 1), omitting those that contain sums. Therefore, in this instance,  $df = (2-1) \times (2-1)$

= 1. In this study used Chi – Square to find out the association between gender and most preferable marketing in advertising

TABLE I. GENDER OF THE RESPONDENTS \* MOST PREFERABLE MARKETING IN ADVERTISING CROSS TABULATION

|                           |        | Most Preferable Marketing In Advertising |                   | Total |
|---------------------------|--------|------------------------------------------|-------------------|-------|
|                           |        | Traditional Marketing                    | Digital Marketing |       |
| Gender Of The Respondents | Male   | 12                                       | 55                | 67    |
|                           | Female | 8                                        | 35                | 43    |
| Total                     |        | 20                                       | 90                | 110   |

TABLE II. RESULT OF CHI-SQUARE

|                                    | Value             | df | Asymp. Sig. (2-sided) |
|------------------------------------|-------------------|----|-----------------------|
| Pearson Chi-Square                 | .008 <sup>a</sup> | 1  | .927                  |
| Continuity Correction <sup>b</sup> | .000              | 1  | 1.000                 |
| Likelihood Ratio                   | .008              | 1  | .927                  |
| Fisher's Exact Test                |                   |    |                       |
| Linear-by-Linear Association       | .008              | 1  | .927                  |
| N of Valid Cases                   | 110               |    |                       |

$H_0$  = There is no significant association between gender and most preferable marketing in advertising.

$H_1$  = There is significant association between gender and most preferable marketing in advertising.

The outcome indicates that the significant value is higher than 0.05, indicating that there is no significant association between gender and the most effective marketing strategy used in advertising.

.P > 0.05.,  $H_0$  is accepted.

### c) Linear Regression:

a) *Regression Coefficient*: The regression coefficient measures how well an Independent Variable (also known as a predictor variable) predicts the Dependent Variable. There are two types of regression coefficients: nonstandard coefficients and standardised coefficients (also called betas). A constant term that predicts the value of the dependent variable is also present. Non-standard coefficients are utilised in equations as coefficients for a variety of independent variables. Standard deviations are used to measure standardised coefficients, though. The

resultant beta value will be the same as the correlation coefficient between the independent and dependent variables if there is only one independent variable used to predict the dependent variable.

*R Values* Regression analysis would provide you with two different R values. A straightforward R value can be used to indicate the correlation between the observed and anticipated values of the dependent variable. The other R values are referred to as R squared. The proportion of the dependent variable's variation that the set of independent variables employed in the model can explain is measured as the R-squared. R-squared is used to assess the independent variable's capacity to predict the dependent variable. However, the R-squared value has a propensity to be greatly inflated when the number of independent variables and the number of instances are both large. The updated coefficient of determination, which provides more specific information about the model's fit, takes these aspects into account. For example, R model's independent variable can predict 70% of the dependent variable's variance. This study used linear regression to find out how gender impacts the convenient of using traditional marketing and digital marketing.

The hypothesis tests if convenient of using traditional marketing & digital marketing impact on gender. To test the Hypothesis H0, the dependent variable convenience of using traditional marketing and digital marketing was regressed on the predictive variable gender. Significant gender-based prediction Convenient,  $F(1,108) = .030$ .

TABLE III. RESULT OF LINEAR REGRESSION

| Hypothesis | Regression Weights  | Beta Coefficient | R <sup>2</sup> | F    | P-Value | Hypotheses Supported |
|------------|---------------------|------------------|----------------|------|---------|----------------------|
| H0         | Gender → Convenient | .042             | .017           | .030 | .863    | YES                  |

H0 = There is no significant relationship between gender and convenient of using traditional marketing & digital marketing.

H1 = There is significant relationship between gender and convenient of using traditional marketing & digital marketing.

The result shows that the significant value is greater than 0.05 which show that there is no significant relationship between gender and convenient of using traditional marketing & digital marketing.

$P > 0.05$ ., H0 is accepted.

## B. Data Analysis

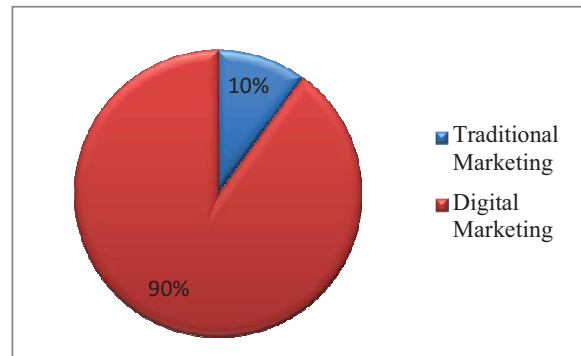


Fig. 1. Marketing which has more effectiveness, feasibility and time saving to the customer.

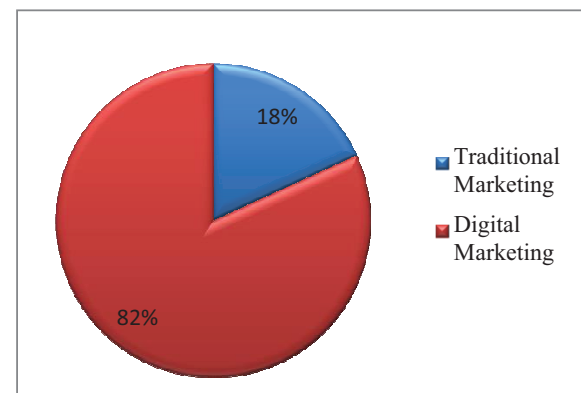


Fig. 2. Marketing which is most preferable in the point of advertising.

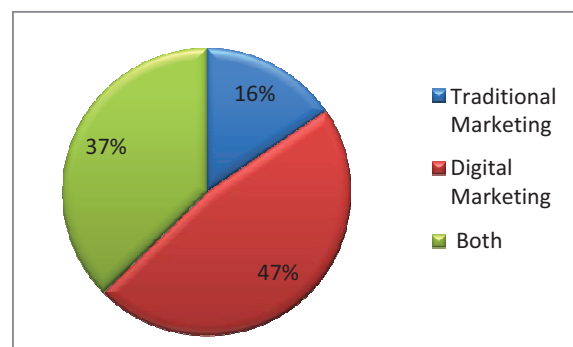


Fig. 3. Marketing which is influencing and attracting the customers more in now a days

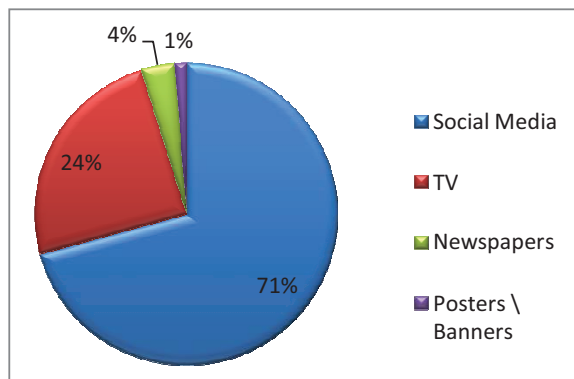


Fig. 4. Medium preferred to watch Ads

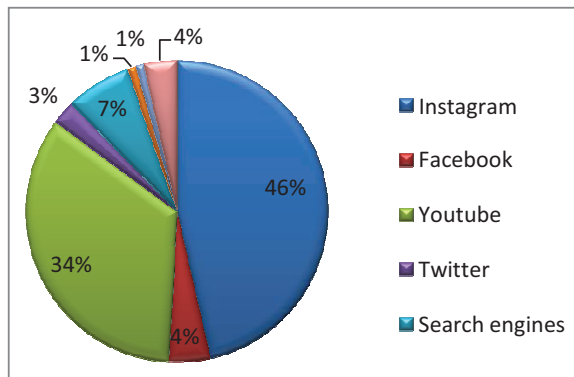


Fig. 5. Digital channels influencing the digital marketing in point of advertising the most

#### IV RESULTS AND DISCUSSIONS

The data determines the satisfaction level of the respondents regarding “An analysis Comparing Traditional Marketing and Digital Marketing (Advertising) using chi square test and linear regression model”. Data were collected on all aspects that help the researcher to find out various findings and suggestions in this study.

- The Male respondents 39% were more than the Female respondents 61%
- The majority of responders, 81%, are between the ages of 21 and 25, while 10.9% are between the ages of 15 and 20.
- Majority of the respondents 56.4% were Under Graduate, 37.3% of the respondents Post Graduate.
- Majority of the respondents were 61.8% Urban and 20.9% Rural
- 37.3% of the respondents income is 20000 to 40000pm and 25.5% of the Respondents income

are between 40001 to 60000

- Majority 82% of the respondents prefer that digital marketing for advertising.
- Most of the respondents (71%) indicated that social media is their preferred medium for displaying ads and 24.5% of them said that TV is preferred medium to watch ads.
- Mostly 46.5% of the respondents said that Instagram is influencing the digital marketing in point of advertising followed by 34.2% YouTube.
- Majority 47% of the respondents said that digital marketing and attracting the customers more in now a days.
- The premise that digital marketing offers significantly greater flexibility of choice than traditional marketing is accepted by no more than 60% of respondents. 53.6% of the respondents agreed that Advertisement through digital media is more effective than traditional marketing, 64.5% of the respondents agreed that Future era will be digital marketing and 63% of the respondents concurred that digital marketing is more convenient than traditional marketing.
- Majority 90% of respondents said that customers can save more time, money, and effort with digital marketing.

#### V SUGGESTIONS OF THE STUDY

- The Companies need to clear with what platform is right for their business.
- The firms should concentrate more on digital marketing because majority of respondents accepts that future era will be full of digital marketing.
- The mix of marketing techniques allows you to attract customers of all ages.

#### VI CONCLUSION

This study was specifically designed to demonstrate how effective and efficient digital marketing is in comparison to traditional marketing. The sector's current marketing strategies were looked into. Data on respondents' opinions of traditional and digital marketing are gathered as part of the survey. The aforementioned research shows that in terms of reach, adaptability, efficiency, and efficacy, digital marketing is superior to traditional marketing. The survey found that consumers prefer digital marketing over traditional marketing when it comes to advertising, and that digital marketing is currently having a greater impact on consumers than traditional marketing.

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